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ASSIGNMENT 01
Business Intelligence & Database Applications

DB MODELLING

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1. Introduction

The nature of a multinational company like Procter and Gamble makes databases, processes and operations complex. This brings, as a consequence, the necessity for a well structured, normalised database that can efficiently organize data while being future proof to any change required.

This requirement does not hold for the current setup as it presents deficiencies in the streamline process of entity relationships. There are four main problems that can be identified, whose identification process will be later detailed in the logical model.

The first one is caused by the many to many relationship between Key Account Managers (KAM) and retailers, meaning that a single KAM can manage multiple clients, and a really big retailer could require many KAM. As a consequence a direct key database would not be able to store this multiple relationships and thus the only options would be to convolute the whole database with repeating entities or leave some empty spaces. This also would make it so a change or removal of a KAM makes the process tedious as there would be a need to change every single one. And with a large enough database this would be really costly and time consuming.

Second we have data duplication in the employee sector as there are multiple entities which use the same employee characteristic, thus creating unnecessary entities which contain the same data type. This makes it so a change in one employee entity can be made for all roles which the employee takes on, leading to inefficient databases. The second consequence is that when using the employees characteristics described in the entity, there would be a need to consult every entity with these details instead of consulting just one which would make the data less accessible.

Third, there is a problem with interdependencies with pricing caused by using the same entity for product hierarchy data and product pricing, this makes it vulnerable to failure in the case of a price change as data is stored in the same part.

Last but important, considering how important customer loyalty is to P&G, is the unstructured loyalty program which makes it difficult to realize customer allocation and studies. If the connection between an invoice and the customer who made the purchase is not

structured as a relationship, the customer may appear as several different consumers creating a distortion in statistical studies.

Some assumptions are, truckers are external contractors so no relationship with the Employee is needed. The Product category is different from the Store Product Category as the Store keeps its own different product categories we assume then that it is different and independent from the product itself.

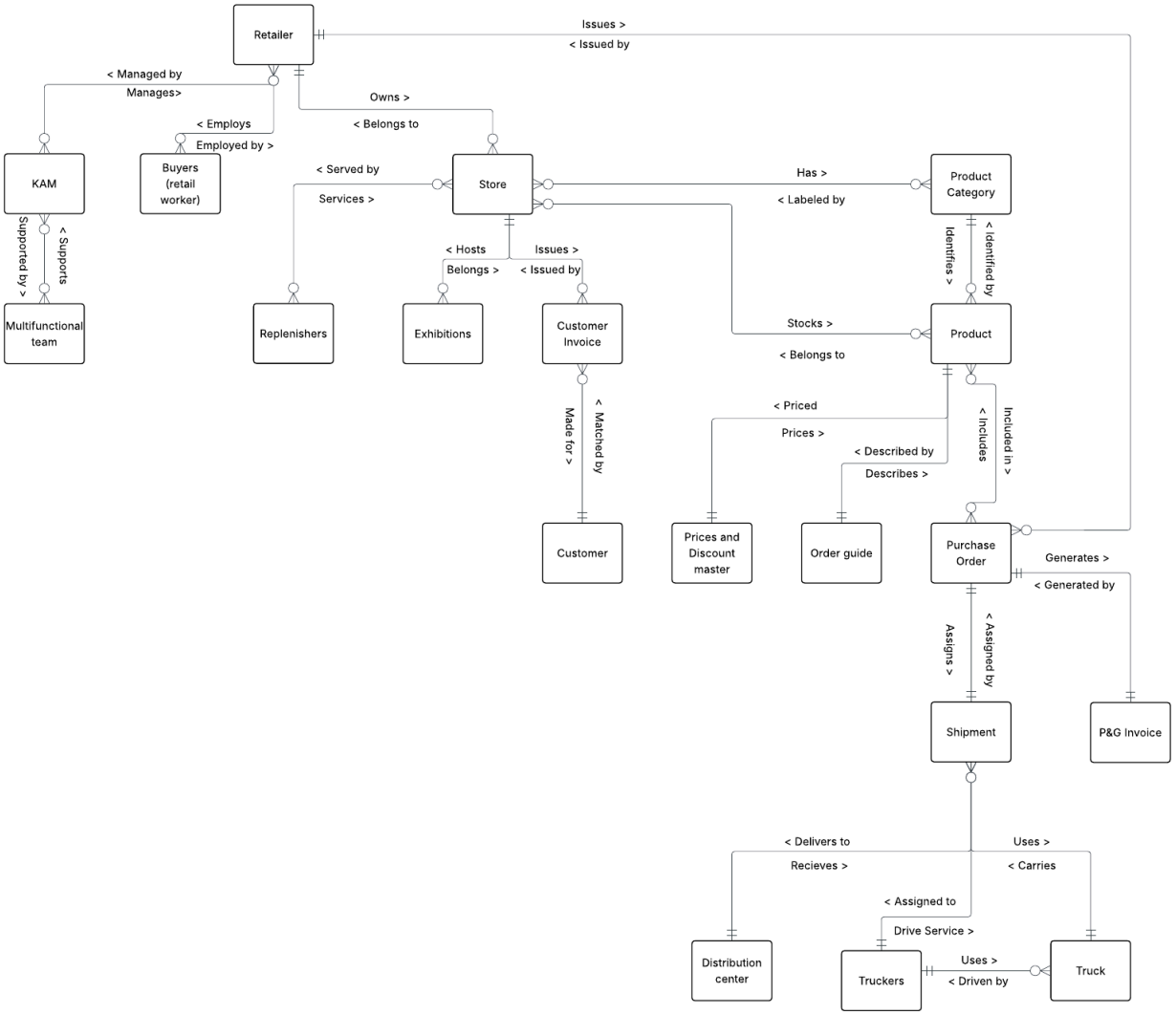
2. Conceptual entity relationship model

Before we can establish a conceptual entity relationship diagram, identification of the entities is necessary. Analyzing the P&G case we identify Retailer, KAM and multifunctional team all part of the main client relationships. In the stores section we can see the entities stores, customers and products as well as replenishers and exhibitions. The products include product, product categories, order guide, prices and discount and invoices. Lastly the purchase order section includes purchase order, shipment, trucks, truckers and distribution centers. All of these are represented in the conceptual diagram as well as in the logical diagram, with the difference that the logical diagram has more entities as it is normalised.

Using these entities the conceptual diagram is built upon the direct relationships mentioned in the case, capturing the essence of the relationships in between entities and how they interact. For example Retailers and KAM have a many to many relationship as described in the case where KAM can manage multiple retailers, and larger retailers may require multiple KAM. KAM and their team is also many to many which means that each KAM is supported by a team, and at the same time the team can assist multiple KAM.

The many to many relationships is the most common, but cases where one to many and one to one are also present. The first case is identified in the retailer to store relationship, one retailer can have multiple stores, but one store is owned by only one retailer. The one to one relationship can be seen especially in the product entity where one product has one order guide only, same applies the other way around. This is in the nature of the entity as a description of a certain product is inherently unique.

With this in mind, the conceptual model reflects the structure of the business as described in the case, without changing the way it behaves in a database. It is just a representation of how the business is ordered.


Retailer
Issues >
< Issued by
< Managed by
Manages>
Owns >
< Belongs to
< Employs
Employed by >
KAM
Buyers (retail worker)
< Served by
Services >
Store
Has >
< Labeled by
Product Category
Supported by >
< Supports
< Hosts
Belongs >
Issues >
< Issued by
Identifies >
< Identified by
Multifunctional team
Replenishers
Exhibitions
Customer Invoice
Stocks >
< Belongs to
Product
Made for >
< Matched by
< Priced
Prices >
< Described by
Describes >
< Includes
Included in >
Customer
Prices and Discount master
Order guide
Purchase Order
Generates >
< Generated by
Assigns >
< Assigned by
Shipment
P&G Invoice
< Delivers to
Recieves >
Uses >
< Carries
< Assigned to
Drive Service >
Distribution center
Truckers
Uses >
< Driven by
Truck

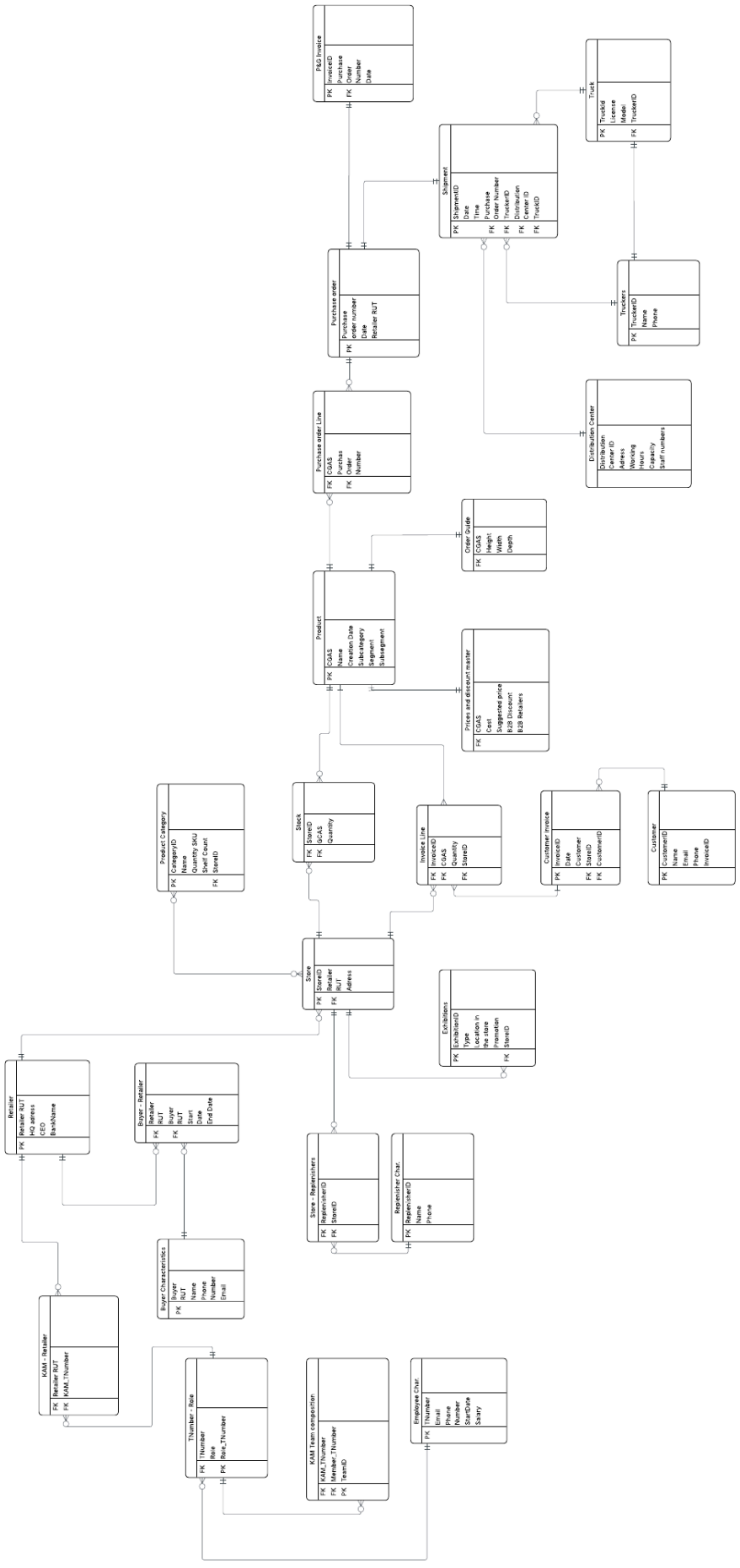
3. Logical entity relationship model

To identify technically the problems that arise, it is important to define what normalisation of a database is and the steps taken to solve this issue. First of all normalisation is the process that ensures that a data model is as efficient as possible ensuring that the relationship of every attribute in an entity to its primary key is correct and direct. Based on this we have the first rule of normalization, which is removing repeating groups in the same entity to two different entities so that it does not interfere with any change. The second rule of normalization is to remove partial dependencies, meaning that every key must depend on the primary key and not just part of it. And last we have the third rule of normalization which is to remove non-key dependencies or transitive dependency where an attribute depends on another attribute rather than on the primary key.

Based on these parameters we can identify the problems. First of all the 1NF is used to fix associative entities, this is applied primarily to the many to many relationships present on the conceptual diagram. This is resolved by creating an entity that can divide and eliminate the repeating groups using foreign keys from both sides.

Second, the 2NF rule checks partial dependencies. An example of this would be the Buyer-Retailer entity whose composite key is formed by BuyerRUT and RetailerRUT while the start and end date attributes depend on both simultaneously as stated on the case, buyers can go from retailer to retailer, therefore one retailer can have more than one buyer and one buyer can work in more than one retailer. In the case this is not fixed P&G would lose the ability to correctly track the specific start and end dates across time.

Third, the 3NF rule is used to remove transitive dependencies, where attributes depend on other non-key attributes rather than the primary key, by separating them. An example of this is the KAM and multifunctional team, in which both use the same characteristics and are at the same time employees. So to avoid repeating characteristics it is split into a different unified entity called employee characteristics which holds universal attributes for describing such employees.


P&G Invoice
InvoiceID
Purchase Order Number
Date
Truck
Truckid
License
Model
TruckerID
Shipment
ShipmentID
Date
Time
Purchase Order Number
TruckerID
Distribution Center ID
TruckID
Purchase order
Purchase order number
Date
Retailer RUT
Truckers
TruckerID
Name
Phone
Purchase order Line
CoAS
Purchas Order Number
Distribution Center
Distribution Center ID
Adress
Working Hours
Capacity
Staff numbers
Order Guide
CoAS
Height
Width
Depth
Product
CoAS
Name
Creation Date
Subcategory
Size
Subsegment
Prices and discount master
CoAS
Cost
Suggested price
B2B Discount
B2B Retailers
Product Category
CategoryID
Name
Quantity SKU
Shelf Count
StoreID
Stock
StoreID
CoAS
Quantity
Invoice Line
InvoiceID
CoAS
Quantity
StoreID
Customer invoice
InvoiceID
Date
Customer
StoreID
CustomerID
Customer
CustomerID
Name
Email
Phone
InvoiceID
Store
StoreID
Retailer RUT
Adress
Exhibitions
ExhibitionID
Type
Location in the store
Promotion
StoreID
Retailer
Retailer RUT
HQ adress
CEO
BankName
Buyer - Retailer
Retailer RUT
Buyer RUT
Start Date
End Date
Store - Repensishers
RepensisherID
StoreID
Repensisher Char.
RepensisherID
Name
Phone
Buyer Characteristics
Buyer RUT
Name
Phone Number
Email
KAM - Retailer
Retailer RUT
KAM.TNumber
TNumber - Role
TNumber
Role
Role_TNumber
KAM Team composition
KAM_TNumber
Member_TNumber
TeamID
Employee Char.
TNumber
Email
Phone Number
StartDate
Sasary

4. Conclusion

The proposed use of both the conceptual and logical entity relationship model provides a solution for the problems identified in terms of structure. It addresses the deficiencies present in the case and helps stabilize the large amount of information contained in the database making it change and future proof. This model is the most adequate because it follows the standardized normalization procedure addressing the many to many relationships.

KAM and retailer relationship was normalised so that it makes it trackable which was not possible before as the separation was non-existent. The unified employee characteristics ensure that data does not repeat and is stored in one easily accessible place, where it can be updated without intervening on other entities. The separation of the price discount master and order guide from the product allows for changes in prices and product characteristics independently from the product itself. Finally the use of loyalty studies is enabled by separating consumer invoice and consumer as it allows for the inclusion of those not subscribed to the loyalty program making the study population complete and not just a subset of it.

The main objective of this project was achieved, making the model less redundant, improving accuracy and efficiency in the use of data and making it reliable especially in handling large amounts of data for a large scale company like P&G.